

SMART INDIA HACKATHON 2026



Problem Statement ID : SIH26073

Problem Statement Title : AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS)

Theme : Disaster Management

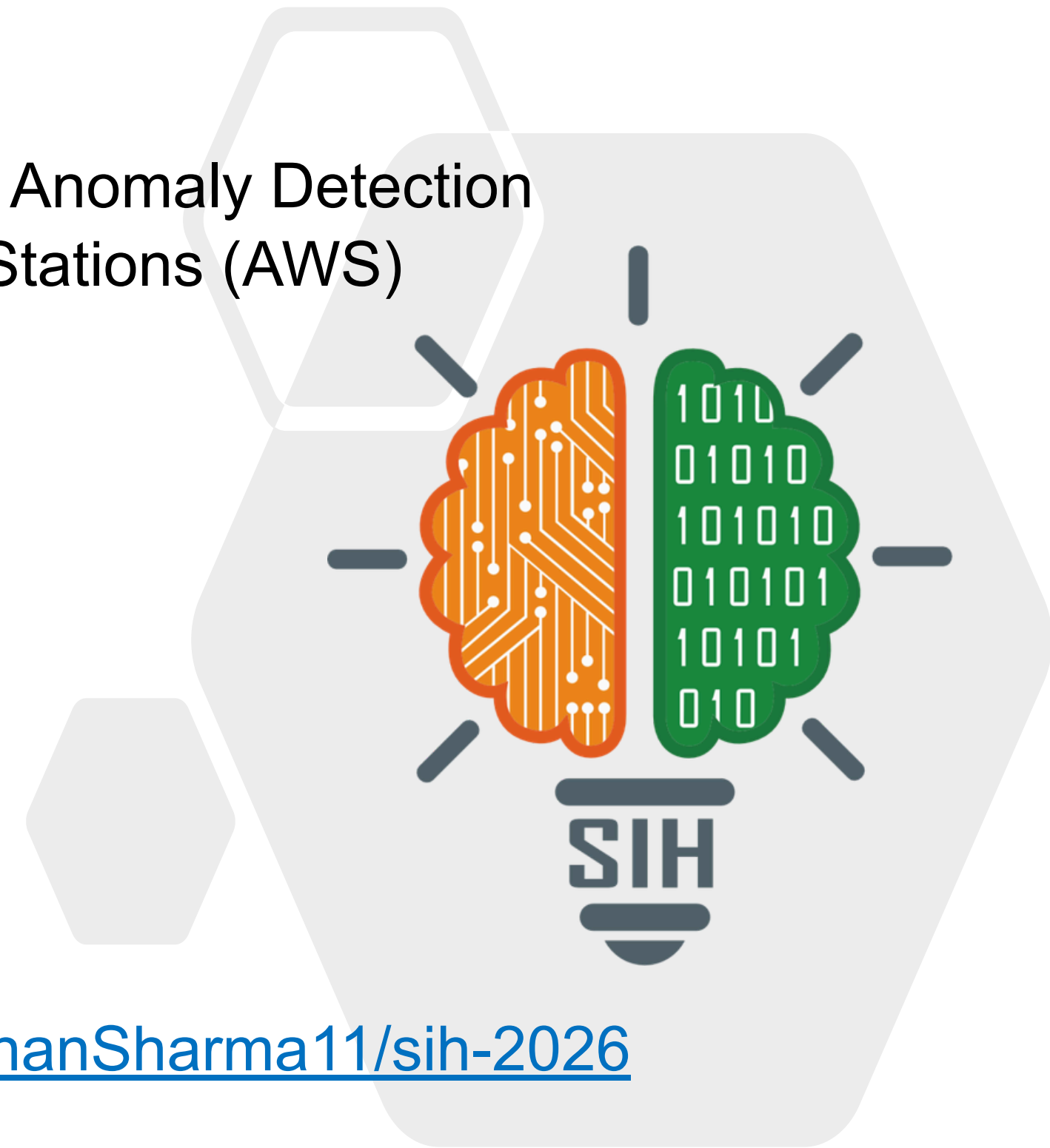
PS Category : Software

Team ID :

Team Name : Think Tank

Github Repo Link : <https://github.com/RoshanSharma11/sih-2026>

Demo Video : <https://shorturl.at/3Z4Ki>



SkyGuard AI – Real-time Quality Control for Indian Automatic Weather Stations (AWS)

Detects anomalies and distinguishes real weather events from sensor faults using only **Temperature, Pressure and Humidity**.

1. THE PROBLEM

AWS data often contains faulty, missing or misleading readings.

**Spikes**

Sudden unrealistic values
(e.g, 55°C)

**Freeze**

Same value for many hours

**Drift**

Slow bias over days

**Comms drop**

Missing data packets

Simple min/max rules cannot tell a real weather event from a broken sensor, leading to false alerts and loss of trust in the weather network.

2. OUR SOLUTION

SkyGuard AI is a real-time quality control service that analyses each hourly reading of **Temperature, Pressure and Humidity** and classifies it as:



Trustworthy
(Clean)



Real Weather Event
(Genuine)



Sensor Fault
(Hardware Anomaly)



Analyses current reading with its recent history and nearby stations



Provides a confidence score, reason and reconstructed value (if needed)



Monitors 7-day station health to keep the network reliable



Runs in real-time for continuous monitoring

3. INNOVATION & UNIQUENESS**Temporal + Spatial Intelligence**

Combines local time-series patterns with nearby stations to distinguish weather events from sensor faults.

**Works with Only T, P, H**

Achieves high reliability using just temperature, pressure and humidity (no external data like wind, radar or satellite).

**Explainable Decisions**

Provides human-readable reason, confidence score and channel-wise contribution.

**Self-Aware, Self-Healing Network**

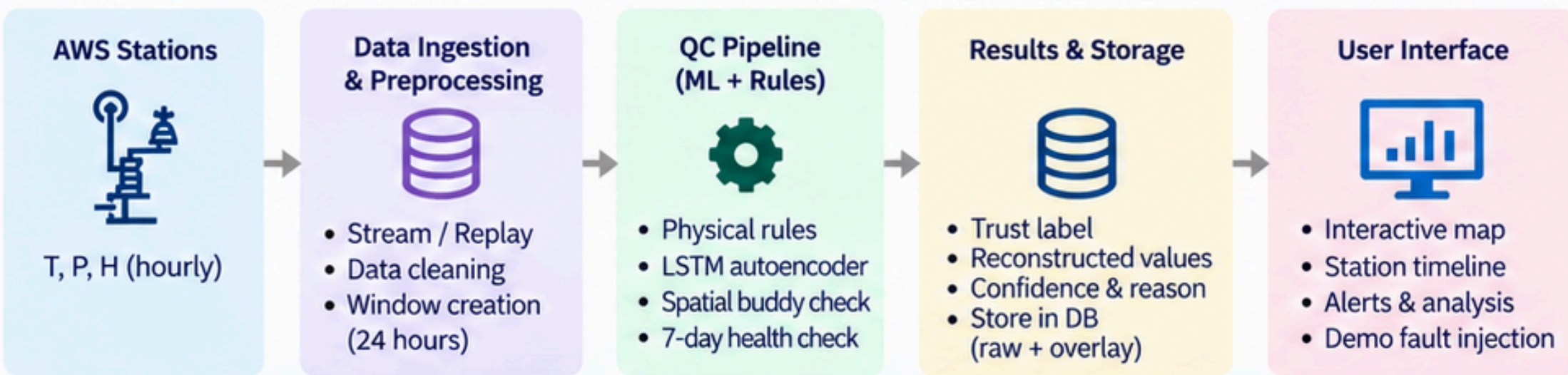
7-day station health monitoring and automatic detection of degraded stations.

**Practical and Scalable**

Built with open-source tools, lightweight and easy to deploy across 100s to 1000s of stations.

A modular, end-to-end pipeline that processes **T, P, H** data in **real-time** to detect anomalies, reconstruct missing/corrupted values, and explain decisions using **temporal + spatial intelligence**.

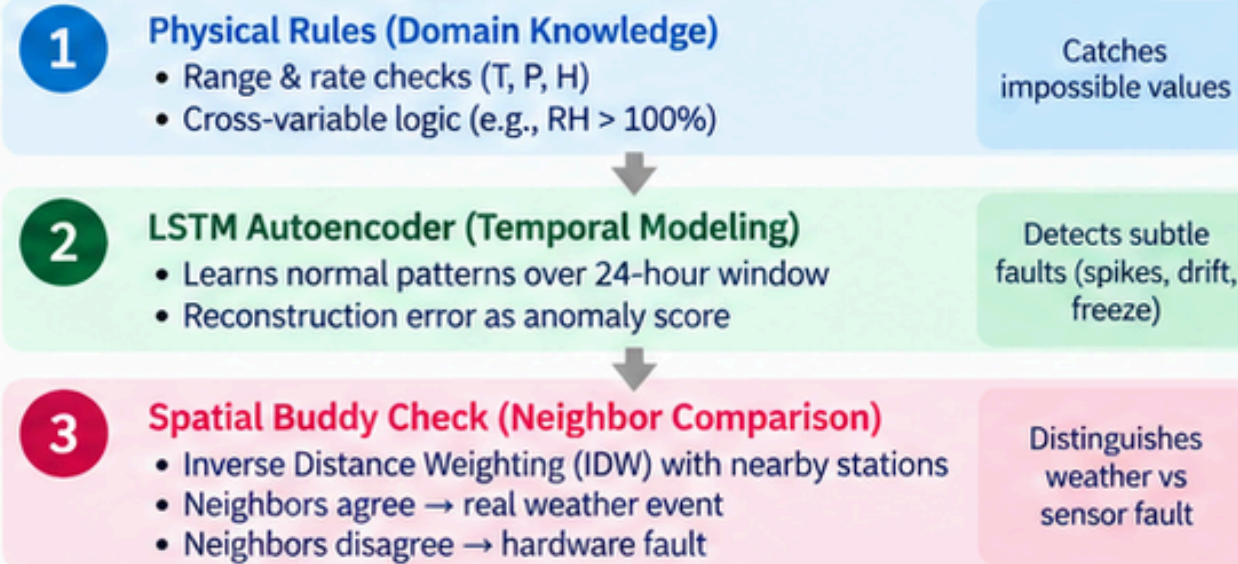
1. Overall System Flow



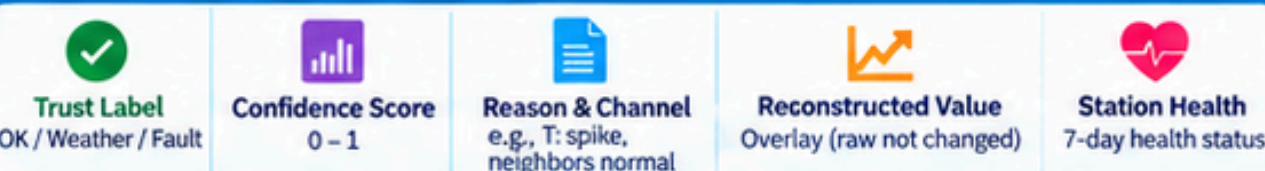
2. Technology Stack

Languages	 Python 3.10+
ML / DL	 PyTorch 
Backend	 FastAPI
Database	 SQLite
Frontend	 Streamlit
Data Processing	 pandas 
Visualization	 Plotly 
Deployment	 Docker 

3. QC Methodology (Detailed)

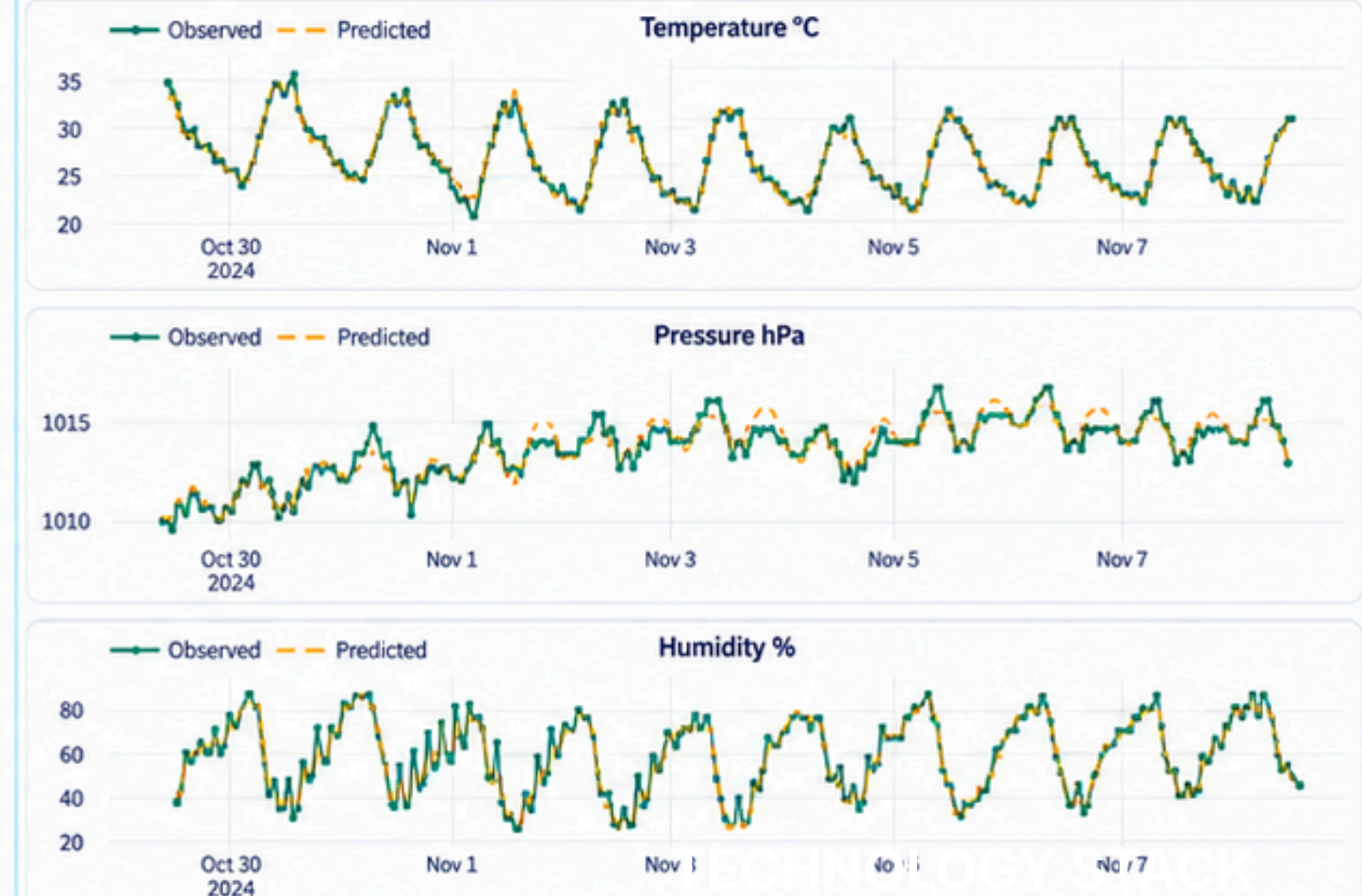


5. Output for Each Hour



4. Example: Reconstruction on Clean Data (Station 42812)

Solid = raw (never overwritten). Dashed = reconstruction overlay.



6. Key Takeaways

- ✓ Uses only T, P, H (no external data)
- ✓ Combines **temporal + spatial intelligence**
- ✓ **Explains every decision** (human-readable)
- ✓ **Scalable and deployable in real-time**
- ✓ Built for a **self-aware, self-healing network**


*From raw data
to reliable
intelligence.*

FEASIBILITY AND VIABILITY

Practical. Scalable. Deployable.

Our solution is **technically feasible**, **cost-effective** and ready for real-world deployment in **Indian AWS networks**.

WHY IT IS FEASIBLE?



Uses existing data

Only T, P, H (available at all AWS stations)



Built with open-source tools

Python, PyTorch, FastAPI, Streamlit



Runs on standard hardware

CPU/GPU (e.g., T4)



Easy to integrate

Works with existing AWS data pipelines

COST-EFFECTIVE

- Open-source stack
- Low infrastructure cost
- Minimal operational overhead



DEPLOYABLE

- On-premise (IMD/State)
- Cloud (NIC / MeitY / private cloud)
- Containerized (Docker)



SCALABLE

- Modular architecture
- Can onboard 100s to 1000s of stations
- Region-wise training



REAL-TIME

- Processes each hour in seconds
- Lightweight models
- Stream-based inference

REAL-WORLD READY

From prototype to national scale



151+
Indian stations trained

1000+
Stations scalable



Target Deployment Pan-India

From pilot to national rollout

KEY CHALLENGES AND RISKS



Missing / noisy data

Sensor gaps, communication drops



Concept drift

Seasonal & regional variations



Infrastructure constraints

Limited resources at remote stations



False positives / negatives

Loss of trust in alerts

OUR MITIGATION STRATEGIES



Physical rules + LSTM reconstruction

Handle missing/abnormal data



Periodic retraining

Season-wise adaptive thresholds



Efficient, containerized deployment

Low compute & memory footprint



Three-tier QC + neighbor comparison

Explainable decisions reduce false alerts

RESOURCE ESTIMATE (Prototype)



Compute

4 vCPU / T4 GPU
(for training & inference)



Storage

~ 10–50 GB
(models + logs)



Memory

8-16 GB RAM



Network

Standard Internet
(HTTP)



VIABILITY OUTCOME



- ✓ Technically feasible
- ✓ Cost-effective
- ✓ Scalable for national deployment
- ✓ Real-time and production-ready
- ✓ Addresses a high-impact problem

A more **trustworthy**, **resilient** and **self-aware** weather data infrastructure for a **safer**, **smarter** and more climate-resilient India.

TARGET AUDIENCE



Government Agencies

(IMD, disaster management, policy makers)

More reliable data for better decision making



Meteorological & Research Institutions

Cleaner data for research and forecasting



Infrastructure & Critical Sectors

Aviation, agriculture, energy, transport, smart cities
Early detection of faulty sensors



General Public

More accurate weather information and warnings
Safer communities

KEY IMPACT AREAS



DISASTER PREPAREDNESS

- Early detection of sensor faults
- Accurate alerts during extreme weather events
- Saves lives and property



AGRICULTURE

- Reliable local weather data
- Better crop planning and irrigation
- Reduces economic losses



AVIATION & TRANSPORT

- Trustworthy weather data for operational safety
- Fewer disruptions due to faulty sensors
- Improves passenger safety



RESEARCH & CLIMATE

- High-quality, cleaned data
- Supports climate research and long-term studies
- Better understanding of extreme weather patterns

**STRONGER
WEATHER INTELLIGENCE
FOR A SAFER INDIA**

SOCIAL, ECONOMIC & ENVIRONMENTAL BENEFITS



Social Impact

- ✓ Safer communities through reliable alerts
- ✓ Supports disaster management
- ✓ Builds trust in public weather services



Economic Impact

- ✓ Reduces losses from false alerts / missed events
- ✓ Improves efficiency in agriculture, aviation, energy and infrastructure
- ✓ Low-cost, scalable solution using open-source tools



Environmental Impact

- ✓ Better monitoring of extreme weather events
- ✓ Supports climate resilience and adaptation
- ✓ Helps in sustainable resource management

BEFORE vs AFTER

Today (Without SkyGuard)

- ✗ Faulty sensors go unnoticed
- ✗ False alerts / missed events
- ✗ Lower trust in data
- ✗ Higher operational risks



With SkyGuard AI

- ✓ Detects sensor faults early
- ✓ Reliable and trustworthy data
- ✓ Improved decision making
- ✓ Safer and more efficient operations

REAL-WORLD VALUE



Higher Safety
for people &
critical infrastructure



**Greater
Economic Efficiency**
across multiple sectors



Climate Resilience
through better data
and monitoring



“Clean, trustworthy weather data is not just a technical improvement — it is a foundation for a safer, more resilient and sustainable India.”

- MeteoStat Python Library: <https://meteostat.net/en/>
- WMO. Guide to Instruments and Methods of Observation (WMO-No. 8).
https://community.wmo.int/en/activity-areas/imop/wmo-no_8
- Ranalkar, M.R. et al. (2015). Operational near real-time QC for Indian AWS. MAUSAM, 66(1).
<https://doi.org/10.54302/mausam.v66i1.370>
- Malhotra, P. et al. (2016). LSTM-based Encoder-Decoder for Multi-sensor Anomaly Detection.
<https://arxiv.org/abs/1607.00148>
- NOAA/NCEP MADIS. Surface QC — spatial consistency (“buddy”) check.
https://madis.ncep.noaa.gov/madis_sfc_qc_notes.shtml